NANYANG TECHNOLOGICAL UNIVERSITY
PH7002 MIDTERM EXAMINATION

Total: 100 points. Time allowed: 90 minutes

Oct 11 2023 (WED)

1. **Landau theory (50 marks)**

   This problem deals with the Landau free energy for an Ising-like order parameter, m, in the absence of an external magnetic field. Assume for this problem that the parameter a varies linearly with T and goes through zero at some temperature T_c, i.e. $a(T) = \alpha(T - T_c)$, and b, c are temperature independent parameters.

   (a) Suppose the free energy has the form

   $$F_1(m) = \frac{a}{2}\, m^2 + \frac{b}{4}\, m^4, \quad b > 0.$$

   Find the equilibrium magnetization m for $T < T_c$ and $T > T_c$. What is the critical exponent β for $T < T_c$? (Note: $m \sim |T - T_c|^{\beta}$). Is this a 1st or 2nd order phase transition?

   (20 marks)

   (b) Suppose instead that the free energy has the form

   $$F_2(m) = \frac{a}{2}\, m^2 + \frac{c}{6}\, m^6, \quad c > 0.$$

   What is the critical exponent β in this case?

   (10 marks)

   (c) Finally, consider

   $$F_3(m) = \frac{a}{2}\, m^2 + \frac{b}{4}\, m^4 + \frac{c}{6}\, m^6, \quad c > 0.$$

   (i) Solve for the equilibrium magnetization m for $b > 0$. Show that there is only 1 real root when $a > 0$ and 3 real roots when $a < 0$. Explain why the transition is continuous (second order).

   (10 marks)

   (ii) Repeat the calculation for $b < 0$. In this case, show that there are 3 real roots when $a < 0$ and 5 real roots when $a > 0$. Sketch $F_3(m)$ as a function of m when $b = -4(ca/3)^{1/2}$ and $a > 0$, showing in particular the values of $F_3(m)$ at the 5 real roots. Hence explain why there is a first order phase transition when $b = -4(ca/3)^{1/2}$.

   (20 marks)

2. **Ground state of spin model (20 marks)**

Find the ground state (stable configuration at $T = 0$) of the one dimensional Ising model with first and second neighbor interactions

$$H = -J_1 \sum_i s_i s_{i+1} - J_2 \sum_i s_i s_{i+2}, \quad s_i = \pm 1.$$

Consider both positive and negative values of the exchange parameters.

(4 marks)

3. **Double Ising Chain (30 marks)**

Consider a double Ising chain such that the nearest-neighbor coupling constant along either chain is J_1 while the one linking adjacent spins in the two chains is J_2, i.e. the lattice is shaped like a ladder, the rungs have coupling J_2 and everything else has coupling J_1. Then, in the absence of the field, the Hamiltonian H is given by

$$H\{s, s'\} = -J_1 \sum_i \left(s_i s_{i+1} + s'_i s'_{i+1}\right) - J_2 \sum_i s_i s'_i.$$

s_i and s'_i are spin $\frac{1}{2}$ variables along the 2 chains and take on values ± 1. Define $K_1 = \beta J_1$ and $K_2 = \beta J_2$. Express the Hamiltonian H in a symmetric form by writing the last term as

$$-\frac{1}{2} J_2 \sum_i (s_i s'_i + s_{i+1} s'_{i+1}).$$

and find the transfer matrix

$$\langle s_i s'_i | T | s_{i+1} s'_{i+1} \rangle.$$

What is T? What is the size of this matrix? Write down all the elements of the transfer matrix explicitly. You do not need to find the eigenvalues.

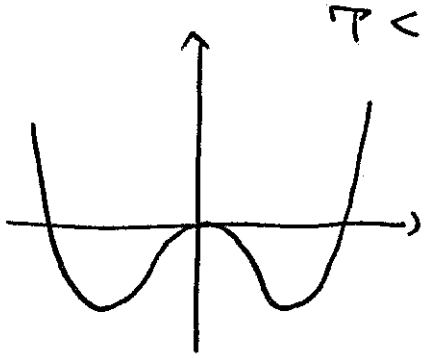
2) <u>Landau</u>

$$F(m) = \frac{a(T)}{2} m^2 + \frac{b}{4} m^4$$

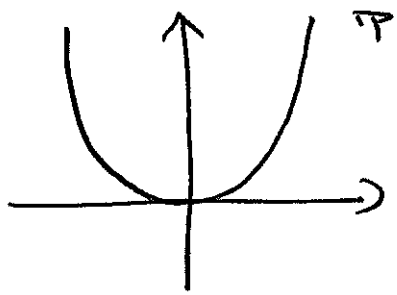
where $a(T) = \alpha(T - T_c)$

$\alpha, b > 0$

$T < T_c$



$T > T_c$



$\underbrace{T > T_c}_{a = \alpha(T - T_c) > 0}$ $\Rightarrow \frac{dF}{dm} = am + bm^3 = 0$

$m[a + bm^2] = 0$

$\Rightarrow m = 0$

$\underbrace{T < T_c}_{a < 0}$ $\Rightarrow \frac{dF}{dm} = am + bm^3 = 0$

$\Rightarrow m = 0$ or $\pm\sqrt{-\frac{a}{b}}$

$$m \simeq \pm\sqrt{-\frac{\alpha}{b}(T - T_c)} \sim [T - T_c]^{1/2}$$

$\Rightarrow \beta = 1/2$

2nd order phase transition.

The solution grows continuously from $m = 0$ at T_c.

b) $F = \frac{a}{2} m^2 + \frac{c}{6} m^6 \quad , c > 0$

$$\frac{\partial F}{\partial m} = 0 = a m + c m^5$$

$$\Rightarrow \quad m = 0 \quad \text{or} \quad m^4 = -\frac{a}{c}$$

$$\Rightarrow \quad m = \left[\frac{\alpha}{c} (T_c - T) \right]^{1/4}$$

$$\Rightarrow \quad \beta = 1/4$$

c) i) $F = \frac{a}{2}m^2 + \frac{b}{4}m^4 + \frac{c}{6}m^6 \qquad c>0$

$b>0 \qquad \frac{\partial F}{\partial m} = 0 = am + bm^3 + cm^5 = 0$

$\Rightarrow \quad m=0 \quad$ or $\quad cm^4 + bm^2 + a = 0$

$$m^2 = \frac{-b \pm \sqrt{b^2-4ac}}{2c}$$

$$\Rightarrow \quad m = \pm\sqrt{\frac{-b \pm \sqrt{b^2-4ac}}{2c}}$$

Since $m^2 \geq 0$, $b, c > 0$,

for $a>0$, $m=0$ is only <u>solution</u>

for $a<0$, $m=0$ and

$m = \pm m_0 \quad \Leftarrow \quad m^2 = \frac{\sqrt{b^2-4ac}-b}{2c} = m_0^2$ } 3 solutions.

When $a=0$, $m^2 = \frac{\sqrt{b^2}-b}{2c} = 0$

so solution grows continuously from 0

$\Rightarrow$ 2nd order (continuous) phase transition.

Yeomans 3.1

↳ related to ANNNI model (axial next nearest neighbor Ising)

see J. Appl. Phys 52(3) 1981

PRB 23 (1) 1981

If $\mathcal{H}_1 = -J_1 \sum_i S_i S_{i+1}$

$J_1 > 0$: $\cdots\uparrow\uparrow\uparrow\uparrow\cdots$ / $\cdots\downarrow\downarrow\downarrow\downarrow\cdots$ ferro (A)

$J_1 < 0$: $\cdots\uparrow\downarrow\uparrow\downarrow\cdots$ Anti-ferro (AF) (B)

If $\mathcal{H}_2 = -J_2 \sum_i S_i S_{i+2}$

$J_2 > 0$: $\cdots\uparrow\uparrow\uparrow\uparrow\cdots$ / $\cdots\downarrow\downarrow\downarrow\downarrow\cdots$ ferro (A)

$\cdots\uparrow\downarrow\uparrow\downarrow\uparrow\downarrow\cdots$ antiferro (B)

$J_2 < 0$: $\cdots\uparrow\uparrow\downarrow\downarrow\uparrow\uparrow\cdots$ (2,2) or $\langle 2 \rangle$ "antiphase" (C)

$$\mathcal{H} = \mathcal{H}_1 + \mathcal{H}_2$$

$$\Rightarrow \mathcal{H} = -J_1 \sum_i S_i S_{i+1} - J_2 \sum_i S_i S_{i+2}, \quad S_i = \pm 1$$

(1) $J_1, J_2 > 0$, (2) $J_1, J_2 < 0$, (3) $J_1 > 0, J_2 < 0$, (4) $J_1 < 0, J_2 > 0$

$E_A = -|J_1| \cdot N - |J_2| \cdot N$ min $\cdots\uparrow\uparrow\uparrow\cdots$ / $\cdots\downarrow\downarrow\downarrow\cdots$ ferromagnetic

(1) $J_1, J_2 > 0$: $\Rightarrow$ (A)

(2) $J_1, J_2 < 0$:

$\cdots\uparrow\downarrow\uparrow\downarrow\uparrow\downarrow\cdots$

$E_B = +|J_1|(-N) + |J_2| N = -N|J_1| + N|J_2|$

$\cdots\uparrow\uparrow\downarrow\downarrow\uparrow\uparrow\cdots$

$E_C = +|J_1| \cdot 0 + |J_2|(-N) = -N|J_2|$

$E_B = E_C$

$\Rightarrow -N|J_1| + N|J_2| = -N|J_2|$

$\Rightarrow |J_1| = 2|J_2|$

$\therefore$ If $|J_1| > 2|J_2|$ then (B), else (C)

(3) $J_1 > 0, J_2 < 0$

$\cdots\uparrow\uparrow\uparrow\uparrow\cdots$

$E_A = -N|J_1| + N|J_2|$

$\cdots\uparrow\uparrow\downarrow\downarrow\uparrow\uparrow\downarrow\downarrow\cdots$

$E_C = 0 - N|J_2|$

$E_A = E_C$ when $|J_1| = 4|J_2|$

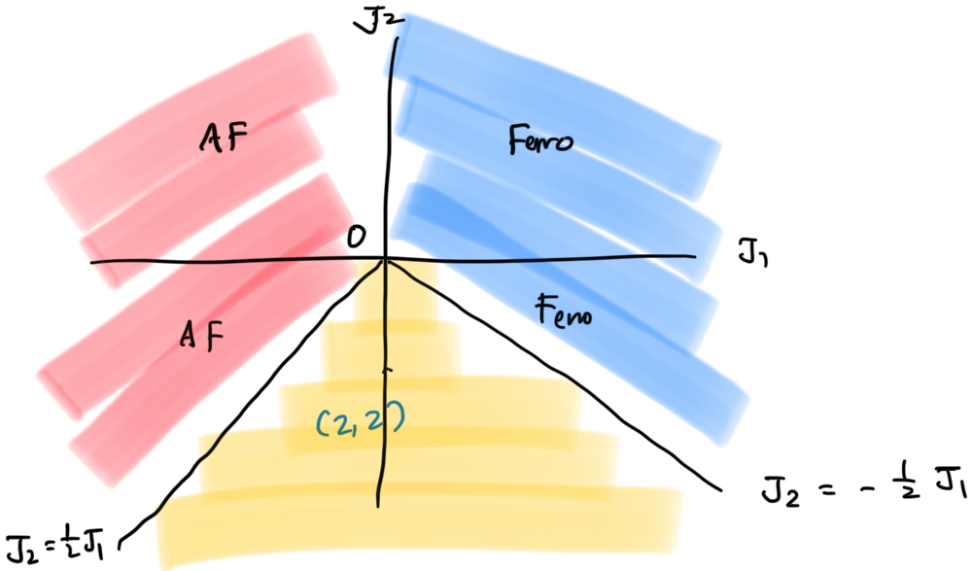
$\therefore$ If $|J_1| > 2|J_2|$ then (A), else (C)

④ $J_1 < 0,\ J_2 > 0$: $\cdots \uparrow \downarrow \uparrow \downarrow \cdots$

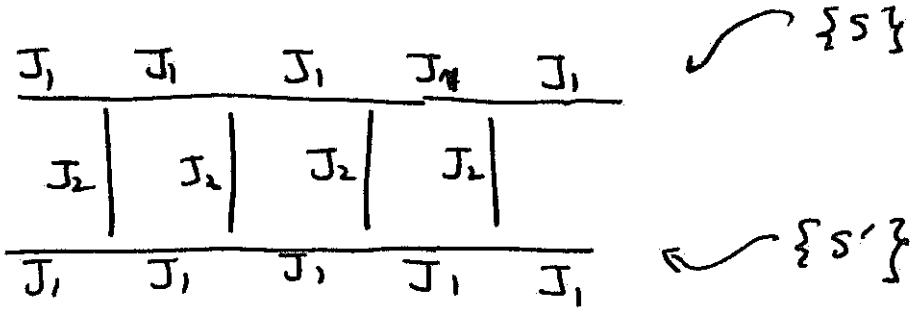
$$E_B = -N|J_1| - N|J_2| \quad (\text{min})$$

anti-ferromagnetic (AF)

Ⓑ always favored.



3)



$$\mathcal{H} = -J_1 \sum_i \left(s_i s_{i+1} + s_i' s_{i+1}'\right) - J_2 \sum_i s_i \cdot s_i'$$

$$\Rightarrow \quad H = -J_1 \sum_i \left(s_i s_{i+1} + s_i' s_{i+1}'\right) - \frac{1}{2} J_2 \sum_i \left(s_i s_i' + s_{i+1} s_{i+1}'\right)$$

$$Z = \mathrm{Tr}\, e^{-\beta H}$$

$$\therefore Z = \sum_{\{s\}}{}' \sum_{\{s'\}}{}' e^{K_1 \sum_i (s_i s_{i+1} + s_i' s_{i+1}') + \frac{K_2}{2} \sum_i (s_i s_i' + s_{i+1} s_{i+1}')}$$

$$= \sum_{\{s\}}{}' \sum_{\{s'\}}{}' \langle s_1 s_1' | T | s_2 s_2' \rangle \langle s_2 s_2' | T | s_3 s_3' \rangle \cdots\cdots \langle T | s_N s_N' \rangle$$

$$T = e^{K_1 (s_i s_{i+1} + s_i' s_{i+1}') + K_2 (s_i s_i' + s_{i+1} s_{i+1}')}$$

$$\therefore \langle s_i s_i' | T | s_{i+1} s_{i+1}' \rangle =$$

$$\begin{array}{c|cccc} & & & & s_{i+1} s_{i+1}' \\ s_i s_i' & -1\,-1 & -1\,+1 & +1\,-1 & +1\,+1 \\ \hline -1\,-1 & e^{2K_1 + 2K_2} & 1 & 1 & e^{-2K_1 + 2K_2} \\ -1\,+1 & 1 & e^{2K_1 - 2K_2} & e^{-2K_1 - 2K_2} & 1 \\ +1\,-1 & 1 & e^{-2K_1 - 2K_2} & e^{2K_1 - 2K_2} & 1 \\ +1\,+1 & e^{-2K_1 + 2K_2} & 1 & 1 & e^{2K_1 + 2K_2} \end{array}$$

4×4 matrix